Campus Connect Platform

National NGO Connect & PCCOE Campus Facility Directory Enhancement

Comprehensive Technical & Implementation Report · Production Release 1.2.0 · Zero-Dependency Node.js

✓ 13/13 PASSING TESTS
100% Verified Ingestion

616	12	194	228
APPROVED NGOS ACROSS INDIA	CAMPUS BUILDINGS MAPPED	CLASSROOMS, LABS & CABINS	FACULTY CABINS ASSOCIATED

1. Executive Overview & Mission

Campus Connect bridges two vital institutional domains within higher education into a unified web application:

1. **National Social Impact (NGO Connect):** Empowering students, faculty, and academic cells to discover, contact, and support verified voluntary organizations across India, while connecting NGOs with relevant central government schemes and CSR pathways.
2. **Campus Resource Management:** Providing complete navigational transparency across the Pimpri Chinchwad College of Engineering (PCCOE) campus through a multi-tier directory of buildings, floors, laboratories, classrooms, and faculty offices, coupled with collision-proof timetable slot booking.

2. Zero-Dependency System Architecture

The application follows an uncompromising **zero-dependency** engineering architecture running on native **Node.js 22.13+**:

- **Database Engine:** Native `node:sqlite` (`DatabaseSync`) executing synchronous queries in Write-Ahead Logging (`WAL`) mode with foreign key integrity enabled. No native compilation, npm modules, or external databases needed.
- **HTTP Core:** Native `node:http` server with sub-millisecond route dispatching, CSRF enforcement via mandatory JSON content negotiation, and constant-time scrypt password hashing (`node:crypto`).
- **Security & Rate Limiting:** In-memory sliding-window request throttling, session tokens hashed via SHA-256, HTTP-only `SameSite=Lax` cookies, and strict Content Security Policy (`CSP`).

3. Enhanced NGO Connect Directory

The NGO Connect database was expanded through ingestion of the official statutory dataset from the **National Trust (MSJE) Voluntary Organizations (VO) Directory** (80th Board Meeting):

- **Scale & Deduplication:** Parsed **569 unique registered organizations** spanning 28+ states and union territories. Combined with 47 preexisting records, the database houses **616 approved NGOs** with zero duplicates.
- **Verified Organizational Schema:** Strict preservation of organization name, head of NGO, executive designation, complete street address, district/city, state, official mobile, and cleaned email. Missing fields are preserved as empty without fabricating fake URLs or donation links.
- **Multi-Faceted Search & Discovery:** Users can filter by State, dynamic City/District dropdowns, 10 cause domains (**Education, Healthcare, Disability, Child Welfare, Women Empowerment, Livelihood, Animals, Environment, Elderly Care, Food Relief**), and browser geolocation distance sort (**"Near Me"**).
- **Government Schemes Integration:** Dynamic matching of 15 statutory schemes (NGO Darpan, 12A/80G, FCRA, CSR-1, Mission Shakti, AVYAY) mapped directly to NGO profiles.
- **Submission & Review Governance:** Multi-tier submission pipeline with automatic duplicate conflict detection (HTTP 409), input sanitization, and administrative moderation workflow.

NGO Schema & Ingestion Pipeline
Safe, automated ingestion with zero duplicate tolerance.

Normalized Columns: `name`, `head_name`, `head_designation`, `city`, `state`, `address`, `phone`, `email`, `website`, `categories`, `focus`, `source`, `status`. Automated parser (`scripts/parse-pdf-data.py`) handles newline stripping, email normalization, and keyword categorization with 100% reproducible execution.

4. Campus Building & Room Directory

Campus Connect establishes an intuitive, hierarchical navigational model:

Campus → 12 Buildings → Floors → Rooms & Labs → Faculty

12 Registered PCCOE Campus Buildings:
- **Building 1:** PCP Admin Office & Fluid Engineering
- **Building 3:** Boys Hostel, Sports Dept, ART Circle, NSS
- **Building 5:** Trust Office, AIML & IT Dept, MCA, Admin
- **Building 6:** AS&H, Computer, Computer Regional, E&TC, CILL
- **Building 7:** MBA & PBS Management Institute
- **Building 7(A):** Old Girls Hostel
- **Building 8:** Architecture, Central Workshop, Multipurpose Hall
- **Building 9:** Mechanical & Civil Engineering Departments
- **Building 10:** New Girls Hostel
- **Building 11:** New Computer Department
- **Sports Complex:** Indoor Sports Hall & Gymkhana
- **PCP Building:** Wings A, B, C, D and E

Floor-by-Floor Mapping for Building 9: Full organization from Basement (-1) to Floor 5:
- **Basement (-1):** NVH Laboratory (G005 LA)
- **Floor 0:** Programming Labs 1 & 2, Multimedia, Digital Electronics, Software Development, Computer Centre (9006 CM), UTM Room
- **Floor 1:** Civil HOD (9104 HO), Mech HOD (9114 HO), Data Structures (9102 LA), Fluid Mechanics, Heat Transfer, Refrigeration
- **Floor 2:** Transportation, Geotechnical, Environmental, Model-Based Systems, Theory of Machines, Dynamics of Machinery
- **Floor 3:** Surveying Lab, Girls Common Room (9314 TG), Elements of Mechanical, Metallurgy & Material Science
- **Floor 4:** System Engineering, Geology, AutoCAD, Battery Tech, Robotics & Automation, 3D PLM Innovation
- **Floor 5:** Examination Cell (9501 EO), PG Research Labs, Computational Mechanics (9511 CM), PG Design Engineering

5. Official Room Codes & Suffix Taxonomy

Strict preservation of official PCCOE suffixes:

Suffix	Room Type / Facility	Bookable?
LA	Laboratory (e.g. 9102 LA)	Yes
HO	HOD Office (e.g. 9104 HO)	No (Staff)
FC	Faculty Cabin (e.g. 9101 FC)	No (Staff)
CR	Classroom (e.g. 5404 CR)	Yes
DH / SH	Drawing Hall / Seminar Hall	Yes
EO	Exam Cell / Admin Office	No (Restricted)
TG / TB	Girls / Boys Common Room	No (Open)
DL	Department Library Section	No (Walk-in)
CM	Computational Mechanics Lab	Yes
TU	Tutorial / Sports Facility	Yes / No

6. Faculty Cabin Directory Association

- **Direct Entity Relationship:** Ingested **228 faculty members** from official records and linked each member to their assigned room via `faculty.room_id` rather than duplicating room entries.
- **Multi-Occupancy Support:** Shared faculty cabins (e.g., Room 9216 housing 6 faculty, Room 9317 housing 7 faculty) properly associate all occupants to the same physical room code.
- **Interactive Bi-Directional Lookups:** Room detail views display assigned faculty chips with email shortcuts; faculty directory searches provide one-click **"Locate Cabin"** buttons jumping directly to building floor plans.

7. Booking Restrictions & Availability Isolation

- **Identity vs. Availability:** Room existence in the directory does not imply booking availability.
- **Restricted Facility Protection:** Confidential facilities (Exam Strong Rooms, HOD offices, staff lounges, student common rooms) are permanently marked `is_bookable = 0`. The booking API strictly throws HTTP 400 if a booking is attempted.
- **Honest Status Communication:** Rooms without active timetable synchronization explicitly display **"Availability not configured"** rather than incorrectly claiming to be **"Available"**.

8. Global Campus Search Engine

A high-speed unified search API (`/api/campus/search`) searches simultaneously across:
- Buildings (by name, department acronyms, or codes)
- Rooms (by official codes `9104 HO`, numeric codes `9102`, lab names)
- Faculty Members (by full name, department, or cabin number)

9. Automated Verification & Test Results

Test Suite Results: 13 Tests Passed (1.3s runtime)
✓ auth portal guard · ✓ csrf content validation · ✓ public ngo filters · ✓ ngo submission duplicate check (409) · ✓ timetable conflict detection · ✓ staff booking permissions · ✓ camera privacy logs · ✓ static security · ✓ campus directory 12 buildings · ✓ official code suffix preservation · ✓ faculty cabin association · ✓ restricted room booking rejection

10. Execution & Deployment

```
# 1. Run Automated Test Suite (In-Memory DB)
npm test

# 2. Start Application Server (Zero Dependencies)
npm start
# Access Web App at http://127.0.0.1:3000
```

Seeded Accounts: `admin@campus.edu` / `faculty@campus.edu` / `student@campus.edu` (Password: `role123`).